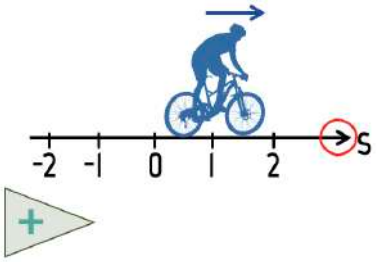
PROFESSOR FERRETTO

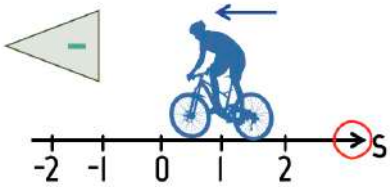
Cinemática - Revisão

Classificação dos movimentos



v + Movimento progressivo

- Progressivo e acelerado (v e a têm o mesmo sinal) — v + a +
- Progressivo e retardado (v e a têm sinais opostos) — v + a -



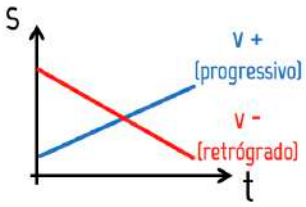
v - Movimento retrógrado

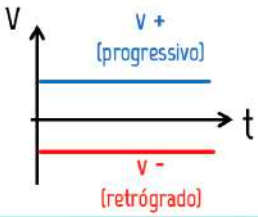
- Retrógrado e acelerado (v e a têm o mesmo sinal) — v - a -
- Retrógrado e retardado (v e a têm sinais opostos) — v - a +

Movimentos

Movimento Uniforme: $v_{constante}$ $(a = 0)$

$s = s_0 + v \cdot t$



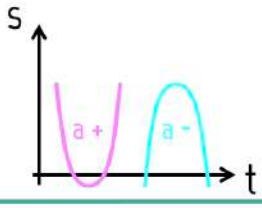


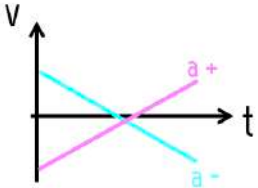
Movimento Uniformemente variado: v_{varia} $(a \neq 0)$

$s = s_0 + v_0 \cdot t + \frac{a \cdot t^2}{2}$

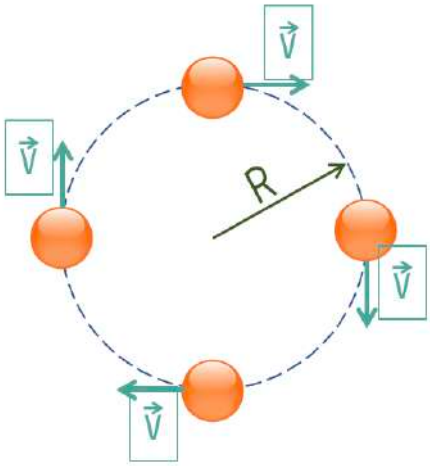
$v = v_0 + a \cdot t$

$v^2 = v_0^2 + 2 \cdot a \cdot \Delta s$





Movimentos circular e uniforme



Período (T): Tempo de 1 volta

Frequência (f): $f = \frac{1}{T}$

velocidade (v): $v = \frac{2\pi . R}{T}$

$v = 2\pi . R . f$

$v = \omega . R$

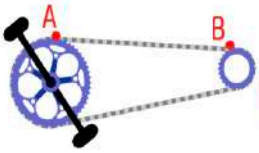
$\omega = \frac{v}{R}$

velocidade angular (ω): $\omega = \frac{2\pi}{T}$

$\omega = 2\pi . f$

A velocidade varia em direção e sentido, porém seu módulo (valor) é constante.

Polias unidas por uma mesma corrente

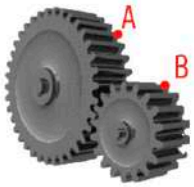


$v_A = v_B$

$\uparrow R_A . f_A = R_B . f_B$

$f_A < f_B$

Engrenagens

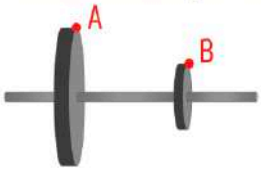


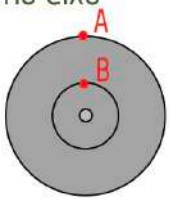
$v_A = v_B$

$\uparrow R_A . f_A = R_B . f_B$

$f_A < f_B$

Polias unidas por um mesmo eixo



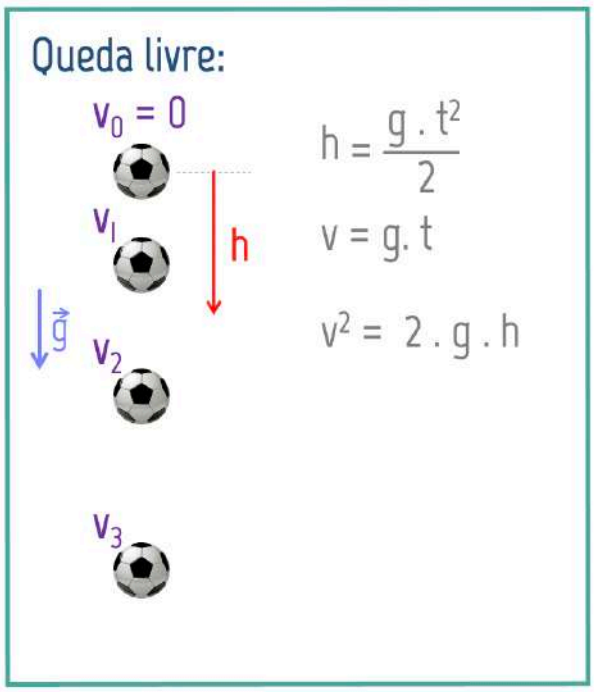


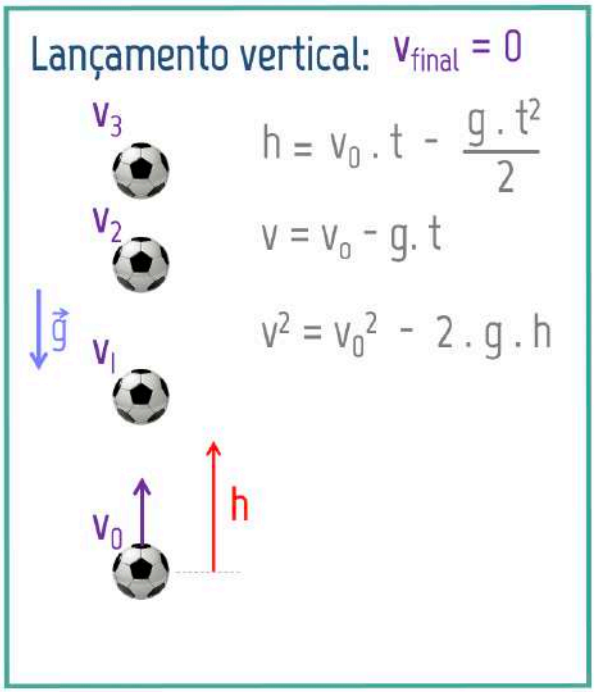
$\omega_A = \omega_B$

$\frac{v_A}{\uparrow R_A} = \frac{v_B}{R_B}$

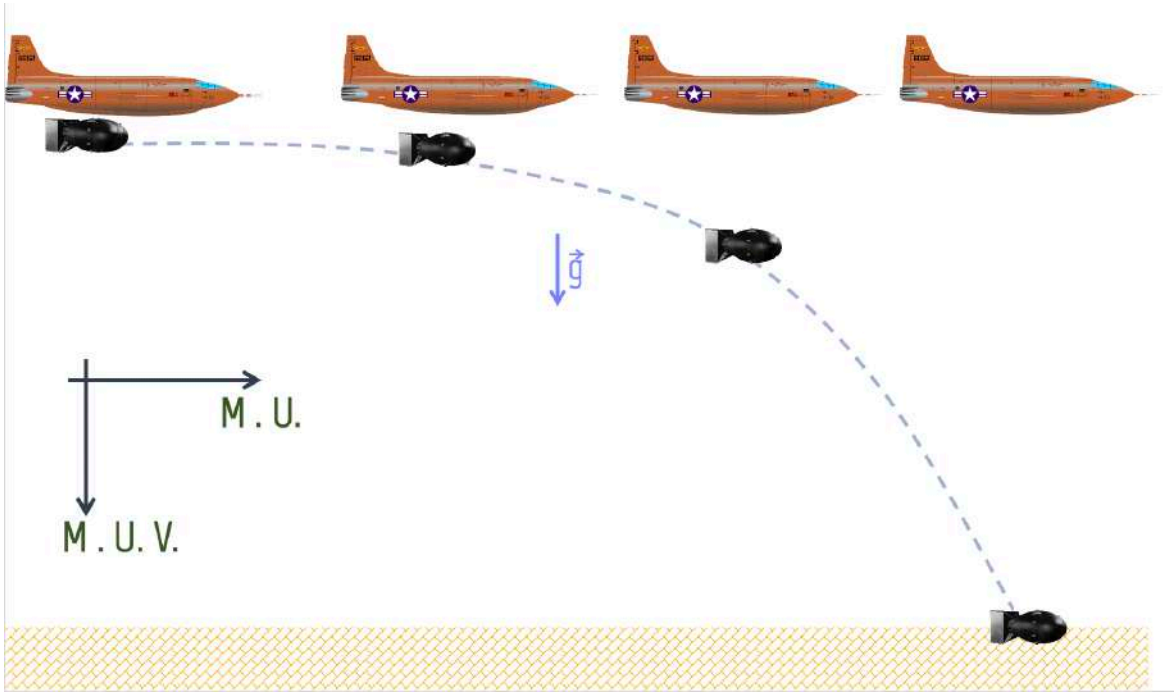
$v_A > v_B$

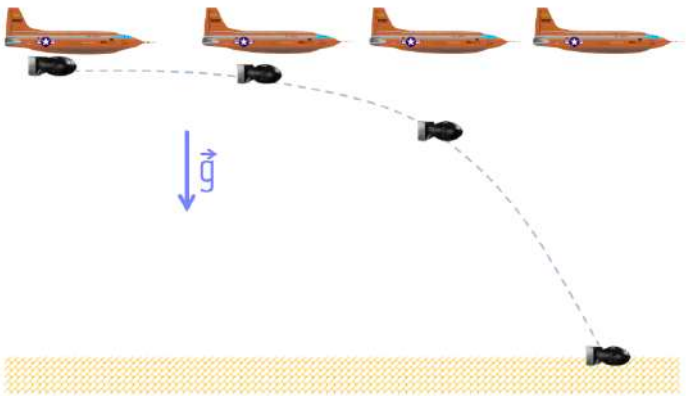
Movimentos verticais

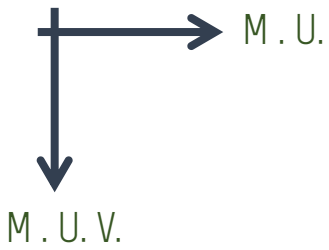




Lançamento horizontal







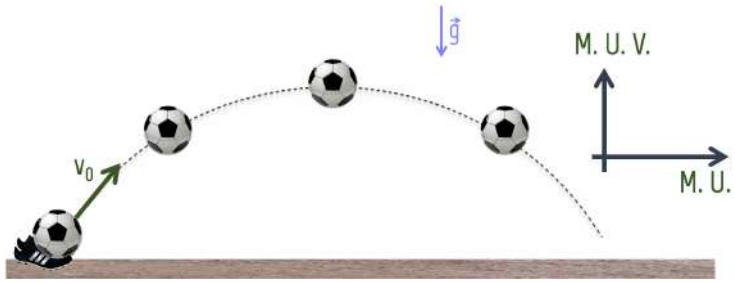
Eixo horizontal:

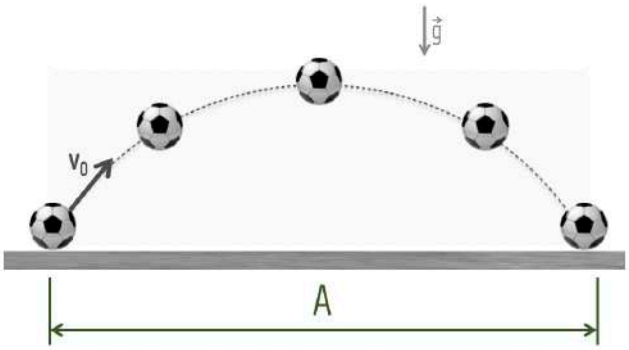
$\Delta s = v_x \cdot t$

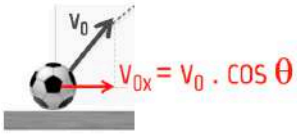
Eixo vertical:

$h = \frac{g \cdot t^2}{2}$ $\quad v_y = g \cdot t$ $\quad v_y^2 = 2 \cdot g \cdot h$

Lançamento oblíquo







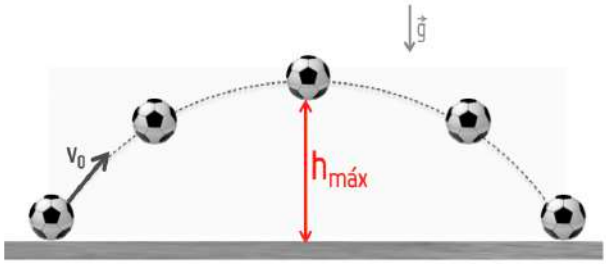
Eixo horizontal:

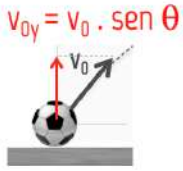
$\Delta s = v_{0x} \cdot t$

Alcance

$A = \frac{v_0^2}{g} \cdot \text{sen } 2\theta$

- Alcance máximo para $\theta = 45°$
- Ângulos complementares geram o mesmo alcance para velocidades iniciais iguais.





Eixo vertical :

$$h = v_{0y} \cdot t - \frac{g \cdot t^2}{2}$$

$$v_y = v_{0y} - g \cdot t$$

$$v_y^2 = v_{0y}^2 - 2 \cdot g \cdot h$$

- na altura máxima $v_y = 0$
- $t_{subida} = t_{descida}$

Anotações: